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## Urban Heat Dynamics in Pune: The Influence of Land Cover and Local Climate

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## Urban Heat Dynamics in Pune: The Influence of Land Cover and Local Climate

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### Abstract

Urban areas with high population density and extensive infrastructure development have been experiencing an increasing strain on the local heat budget, leading to a surge in heat-related illnesses and discomfort. This study examined the impact of climate and land use as heat islands in Pune, India, from 2012 to 2023 at six different locations representing varying degree of urbanization. Satellite land cover observations revealed that 55.17% of the total area was urbanized in the city itself, which was limited to 44.8% in 2012. This urbanization has significantly impacted the increasing tendency of maximum temperature ( $T_{\max}$ ;  $0.13^{\circ}\text{C year}^{-1}$  to  $1.63^{\circ}\text{C year}^{-1}$ ) at almost each study site and minimum temperature ( $T_{\min}$ ;  $0.06^{\circ}\text{C year}^{-1}$  to  $0.23^{\circ}\text{C year}^{-1}$ ) at a specific location during night. The mutual effect of land cover changes and meteorological conditions have evidenced the heat islands with varying intensities ( $2^{\circ}\text{C}$  to  $8^{\circ}\text{C}$ ) at four of the six sites, with significantly intensifying rates from  $0.05^{\circ}\text{C year}^{-1}$  to  $0.39^{\circ}\text{C year}^{-1}$ . The estimation of dominating land cover type for the formation of heat islands demonstrated a significant simple determination ( $r^2 = 0.001$  to  $0.013$ ) and probability ( $P < 7.91 \times 10^{-13}$  to  $2.33 \times 10^{-5}$ ) with heat island temperature identifying urban land cover as the primary factor at two sites, while the other two were affected by mixed land covers influenced by local meteorological characteristics. The outcomes of this study offer valuable insights into the development of heat islands in Pune and could guide strategies for alleviating urban heat, ultimately improving climate resilience and thermal comfort citywide.

**Keywords:** *climate change; heat waves; UHI; urban climate*

## 1. Introduction

Climate change is a primary global concern that necessitates the immediate implementation of mitigation policies. The impacts of climate change are being felt by almost every nation worldwide. Rising average temperatures and extreme weather events are clear manifestations of global warming and climate change [1, 2, 3]. Urban development and human settlements have altered ecosystems and natural stocks, which are directly linked to climate change [4, 5]. The rapid upsurge in population has created densely colonized urban pockets that require more energy consumption [6, 7, 8] and result in higher greenhouse gas (GHG) emissions [9]. Urbanization plays a crucial role in changing the local and regional heat balance [10]. Buildings [11], roads, and urban emissions either absorb, reflect, or trap the radiative energy [12], ensuring higher near-surface temperatures [13]. This phenomenon is observed through the higher temperatures in urban areas compared to the surrounding non-urban regions, known as the urban heat island (UHI). The UHI intensity at a particular time can be calculated using formula 1 below:

$$UHI_h = (T_h - T_{mean}) \quad (1)$$

where  $UHI_h$  is heat island intensity at time  $h$ ,  $T_h$  is the temp of the location at time  $h$ , and  $T_{mean}$  is the mean temperature of non-urban stations at time  $h$ .

The magnitude of heat island in urban environment is profoundly influenced by urban characteristics [14, 15], synoptic conditions [16], local meteorological attributes, composition of urban materials [17, 18], and the presence of green spaces [19, 20]. Various researchers have performed several studies using in-situ and space-based observational data to investigate the cause of heat islands and mitigate them in different urban environments. Ramsay et al. [21] applied Landsat and in-situ surface temperature data to study UHI's spatial and temporal variation in Makassar, a coastal city in Indonesia. They found the UHI intensity up to 9.2°C in urbanized and 6.3°C in non-urban environments. Kasniza Jumari et al. [22] converted infrared radiation from the Landsat satellite into thermal images to identify the land surface temperature in Kuala Lumpur, Malaysia. They discovered the highest temperature difference between urban and rural regions from 10.8°C to 25.5°C in 2013 and from 16.1°C to 26.73°C in 2021. Space-based observations can provide a macro environment to mitigate the UHI due to revisit time and the uncertainty associated with the data. However, it may fill the gap of data scars regions. Dudorova and Belan [23] proposed a quantitative model utilizing the difference in variable radiation fluxes between urban and non-urban settlements in Tomsk, Russia. Their model may not apply to tropical climate systems due to uncertainty in microclimate. Marando et al. [24] simulated the temperature difference between a baseline and a no-vegetation scenario for 601 European cities to report the microclimatic regulation of urban green infrastructure. He concluded that a tree cover of at least 16% is required to achieve a 1°C drop in urban temperatures. In India, due to diverse microclimates across different climate systems, it is challenging to

develop a unified model to categorize the UHIs. Very few studies have been performed in India to investigate the characteristics of UHI in different climate zones. Singh et al. [25] analyzed Landsat data in Bhopal to investigate the correlation between land surface temperature and built-up area. They observed a rise of 4°C in surface UHI temperature in 2020 compared to 2008. Borbora and Das [26] examined the summertime UHI in Guwahati and found a marginal difference of 0.17°C between daytime and nighttime UHI temperatures. The present study is based on Pune, a rapidly growing metro city in Maharashtra, India. Pune is a key hub for the IT, automobile, and manufacturing industries. It is commonly known as the "Oxford of the East" due to its prestigious educational establishments. The city of Pune has consistently been designated as the most liveable city in India; consequently, it has experienced rapid urbanization and human habitation. Very few studies have been conducted about the climate of Pune, resulting in a limited comprehension of the environmental impacts of urbanization. The purpose of the present study is three-fold:

- (a) To investigate how the climate of Pune has evolved in the past 11 years.
- (b) To understand how urbanization has influenced the city's climate.
- (c) What ideal conditions lead the city towards heat islands?

## 2. Methods

### 2.1 Study area

Pune is situated on the western edge of the Deccan plateau, on the sheltered side of the Sahyadri mountain range, which acts as a barrier against the Arabian Sea. Pune has a tropical wet and dry climate, closely bordering on a hot semi-arid climate. Pune witnesses three distinct seasons: summer, monsoon, and winter. The usual summer period lasts from March to mid-June. The month of March witnesses a surge in temperatures due to enhanced sunlight, with daytime peaks reaching 35°C and nighttime lows around 18.9°C. During this time, there are partly to mostly cloudy skies and increased relative humidity (RH) at night (~60%) in comparison to the daytime (~20%). In April, the temperatures escalate even more, with average daytime highs of 37.1°C and lows at 21.7°C, accompanied by similar cloud patterns and a rise in nighttime RH (~67.6%). May emerges as the hottest month, with temperatures soaring up to 42°C and a daytime maximum of 37.1°C. The presence of cloud cover at night results in higher RH (76.2%) and more temperate nighttime temperatures (23°C). Despite the intense heat, nights in Pune tend to be pleasantly cool due to its elevated altitude. Pune has experienced very hasty urbanization and industrial developments, which have impacted its microclimate.

## 2.2 Meteorological observations and pre-processing

The India Meteorological Department (IMD) operates 32 meteorological observational stations in Pune district, with six stations located within the Pune Municipal Corporation (PMC) area (18.43601,73.74396, 18.65404, 73.93455) being used for the current study (Figure 1). These stations represent areas that have undergone rapid urbanization (Chinchwad and Pashan), industrial development (Magarpatta), the city center (Shivajinagar), and relatively less-urban settings (National Defence Academy). Meteorological data from above six stations was employed in the current study at a temporal resolution of 15 minutes from 1st January 2012 to 31st December 2023.

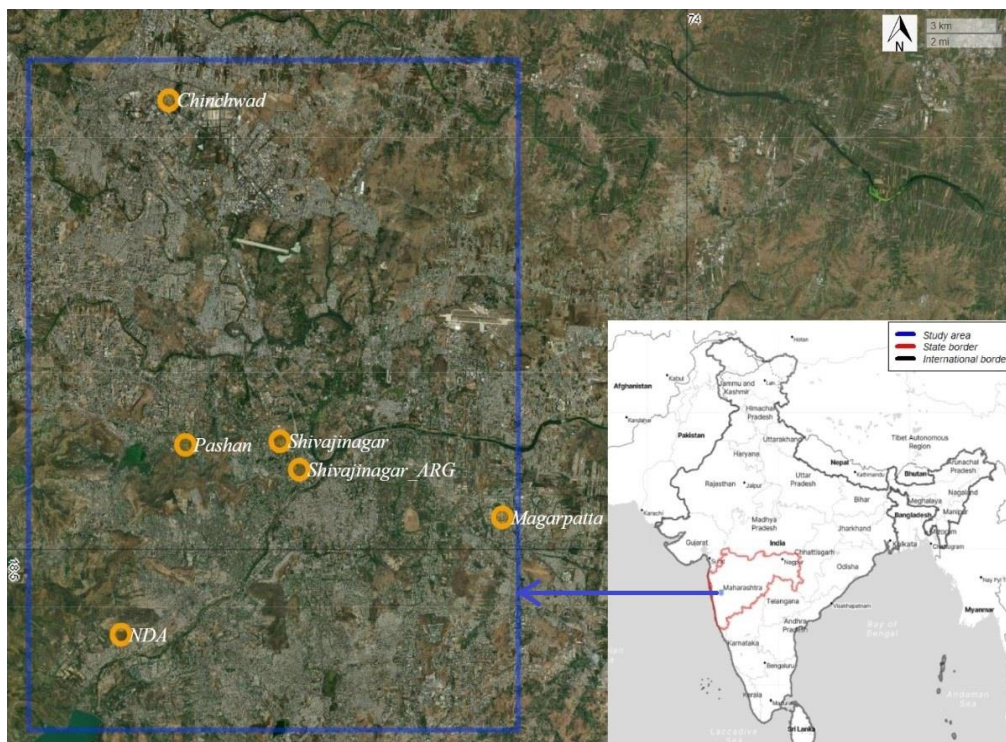


Figure 1. The AWS locations were utilized in our research to evaluate the changes in the microclimate of Pune city. The yellow points on the map indicate the AWS stations, while the blue bounding box outlines the PMC area. In the inset map, the black outline delineates India's international border, the red outline marks the boundary of Maharashtra state in India, and the blue box represents the PMC boundary in Pune.

Analysis was conducted on the diurnal variations of near-surface temperature and relative humidity (RH) during March, April, and May from 1st January 2012 to 31st December 2023, to pinpoint the hours with maximum and minimum values. After examining temperature and RH diurnal variations, specific daytime and nighttime hours were selected as daytime and nighttime threshold hours. UHI intensity was calculated at each location with respect to the National Defence Academy (NDA) station for both daytime and nighttime thresholds. The trend of UHI, maximum temperature ( $T_{\max}$ ), and minimum temperature ( $T_{\min}$ ) at each location during these threshold hours was also evaluated using Theil and Sen's slope estimation (Theil, 1950; Sen, 1968) in the R programming language (<https://www.r-project.org>).

### 2.3 Land use land cover data

Land use land cover (LULC) is one of the most important characteristics while investigating the UHI. Effects of urbanization directly influence the urban climate. European Space Agency (<https://www.esa.int/>) is continuously monitoring the changes in land and atmosphere through the Copernicus program ([https://www.esa.int/Applications/Observing\\_the\\_Earth/Copernicus](https://www.esa.int/Applications/Observing_the_Earth/Copernicus)).

Copernicus's climate change initiative (CCI) is the sub-program that focuses on the climate parameters using the Sentinel satellites. Land cover classification data (<https://cds.climate.copernicus.eu/cdsapp#!/dataset/satellite-land-cover?tab=overview>) by the European Centre for Medium-Range Weather Forecasting (ECMWF) from 2012 to 2022 was analyzed to explore the percentage changes in the urban area within PMC boundary. The LULC data is provided annually at a spatial resolution of 300 meters with distinct land cover classes. The global LULC dataset was cropped to the extent of the study area, and the pixels classified by seven land cover types including cropland, herbaceous cropland, irrigated cropland, mosaic cropland, natural vegetation, grassland, and urban were subsetted from each year. This subsetted LULC data was extracted for a surrounding area of 36 cells ( $\sim 1 \text{ KM}^2$ ) at each station. Further, these extracted land cover classes were modeled with heat island intensity at each station in R programming with *gstat* and *forecast* libraries to examine the predominant land cover for UHI at the individual stations.

## 3. Results and Discussion

### 3.1. Changes in land cover

The comprehensive evaluation of the entire study area indicated a substantial expansion of urban areas by 49.51 square kilometers in 2022 compared to the land cover extent in 2012 (Table 1). The land cover change has highlighted a notable increase in urban areas in Pune, particularly evident in Chinchwad, Magarpatta, and Pashan regions, where there was a decrease in trees and vegetation cover due to ongoing construction and urban development projects

(Figure 2). Vegetation and tree cover are directly associated with the thermal properties of the land surface. Reduced Vegetation and tree cover to increase the built-up area significantly contribute to absorbing more radiative energy. The expanded built-up area in Pune from 2012 to the present due to industrial and residential settlements has influenced the heat balance within different parts of the city.

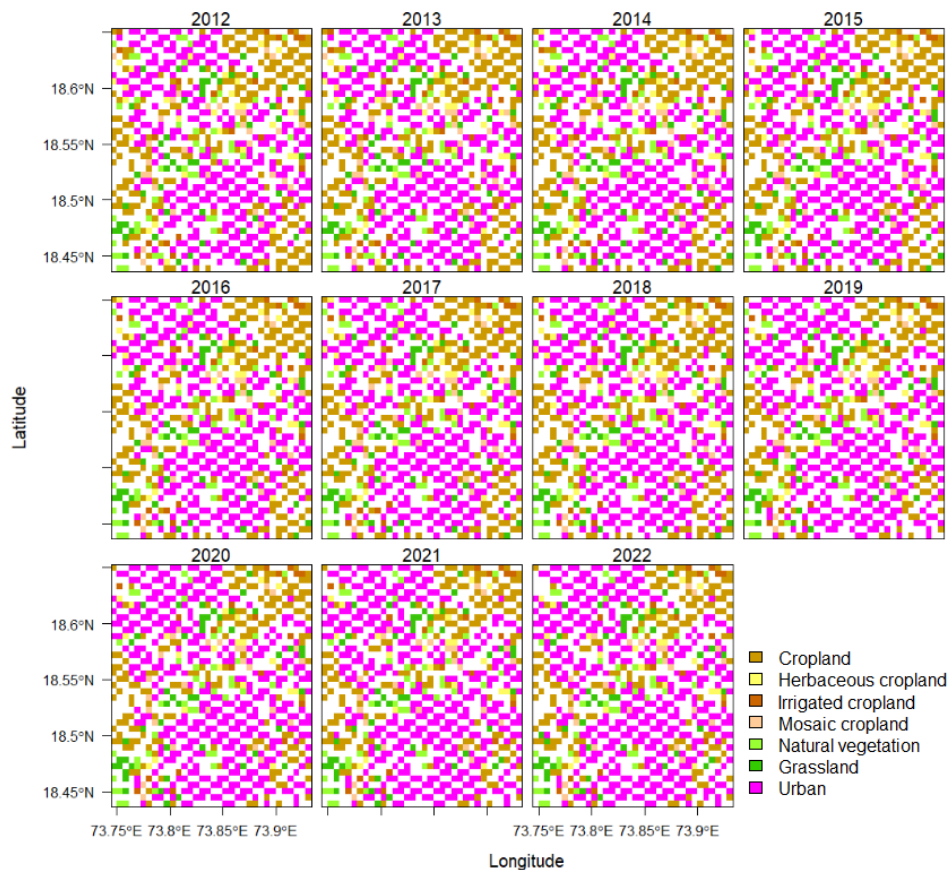


Figure 2. Expansion of urban area under PMC boundary from 2012 to 2022.

Table 1: Urban area expansion in Pune city from 2012 to 2022.

| <i>Year</i> | Pixel counts | Area (KM <sup>2</sup> ) | Proportion (%) |
|-------------|--------------|-------------------------|----------------|
| 2012        | 2379         | 214.11                  | 44.82          |
| 2013        | 2440         | 219.6                   | 45.9           |
| 2014        | 2483         | 223.47                  | 46.7           |
| 2015        | 2497         | 224.73                  | 47.05          |
| 2016        | 2507         | 225.63                  | 47.23          |
| 2017        | 2526         | 227.34                  | 47.59          |
| 2018        | 2580         | 232.2                   | 48.61          |
| 2019        | 2752         | 247.68                  | 51.85          |
| 2020        | 2853         | 256.77                  | 53.75          |
| 2021        | 2884         | 259.56                  | 54.34          |
| 2022        | 2928         | 263.52                  | 55.17          |

### 3.2 Identifying threshold hours

The near-surface temperature was observed maximum during the pre-monsoon season (March, April, and May) at all study sites based on the time variation of average temperature. The increase in diurnal temperature and decrease in diurnal RH from 2012 to 2023 is noticeable after 07 Hrs, reaching its peak between the 15<sup>th</sup> and 16<sup>th</sup> hour of the day. Conversely, a decrease is noted after the 18<sup>th</sup> hour, with the lowest temperature around the 05<sup>th</sup> and 06<sup>th</sup> hours of the day (Figure 3). This pattern differs from most of the metro cities in India, where maximum and minimum temperatures occur around the 14<sup>th</sup> and 02<sup>nd</sup> hour of the day, respectively. In the premonsoon season, dust particles in the atmosphere induce convective cloud formation from late afternoon until evening hours (P.D. Safai et al., 2013), contributing to increased surface air temperature. Utilizing these findings, we identified the daytime (13<sup>th</sup>, 14<sup>th</sup>, 15<sup>th</sup>, and 16<sup>th</sup>) and nighttime (03<sup>rd</sup>, 04<sup>th</sup>, 05<sup>th</sup>, and 06<sup>th</sup>) hours in the premonsoon season as key points for our study.

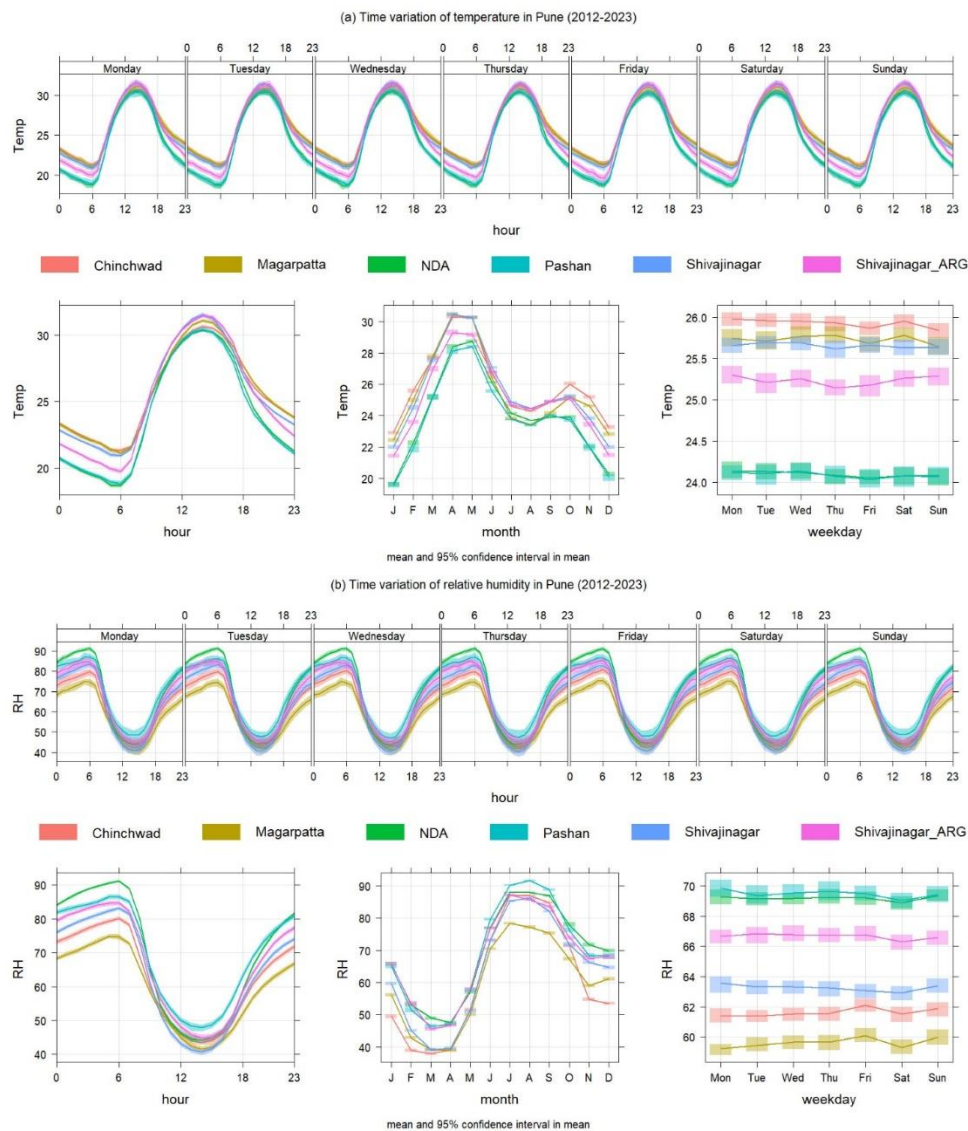


Figure 3. Time variation of temperature (a) and RH (b) at each observational station from 2012 to 2023. The plot shows variation in the hours of the weekdays, diurnal, monthly, and day of the week variation.

### 3.3 Trend of $T_{\max}$ and $T_{\min}$

$T_{\max}$  and  $T_{\min}$  are the primary variables influenced by urbanization, population density, emissions, and particulate aerosols, all collectively exert a discernible influence on the local climate conditions. The analysis reveals an upward trend ( $0.13^{\circ}\text{C year}^{-1}$  to  $1.63^{\circ}\text{C year}^{-1}$  in nighttime temperatures at almost every monitoring station, prompting significant concerns (Table 2). Specifically, locations such as Magarpatta, Shivajinagar, and Pashan have consistently displayed this increasing trend throughout the premonsoon season. Chinchwad has exhibited a similar trend not only during nighttime but also during daytime in April and May. Moreover, a distinct increase in  $T_{\min}$  was observed at Chinchwad in April and May during nighttime, indicating a clear temporal pattern in temperature variations. Chinchwad is known for its automotive, pharmaceutical, electrical, electronics, hardware, aerospace, and manufacturing sectors with ongoing industrial and infrastructural expansions. On the other hand, Magarpatta, situated in the Hadapsar suburb, has evolved into a significant industrial hub characterized by high population density and extensive development, particularly marked by the proliferation of high-rise buildings within its technological park and surrounding areas. These tall structures play a pivotal role in absorbing heat, thereby disrupting the region's natural radiative cooling processes and thermal equilibrium. Similarly, Shivajinagar, the central area of Pune, faces challenges posed by heavy vehicular traffic and rapid urbanization. Recent infrastructural projects such as the metro-rail network have further exacerbated the emission levels and heat-retaining properties of construction materials in the vicinity. Interestingly, the less-urban setting, NDA, has witnessed a significant increase in  $T_{\max}$  values during the nighttime hours of March and May. The increasing human settlements due to lakes and mountains in the area have contributed to this rising trend in  $T_{\max}$ .

Table 2: Trend of  $T_{\max}$  and  $T_{\min}$  in the pre-monsoon season in Pune from 2012 to 2023 at threshold hours. The highlighted text in dark red indicates significantly increasing values.

| The trend of maximum and minimum temperature ( $^{\circ}\text{C year}^{-1}$ ) |    |           |       |       |            |       |       |        |       |       |       |       |       |              |       |       |                  |       |       |
|-------------------------------------------------------------------------------|----|-----------|-------|-------|------------|-------|-------|--------|-------|-------|-------|-------|-------|--------------|-------|-------|------------------|-------|-------|
|                                                                               | Hr | Chinchwad |       |       | Magarpatta |       |       | Pashan |       |       | NDA   |       |       | Shivajinagar |       |       | Shivajinagar_ARG |       |       |
|                                                                               |    | MAR       | APR   | MAY   | MAR        | APR   | MAY   | MAR    | APR   | MAY   | MAR   | APR   | MAY   | MA_R         | APR   | MAY   | MAR              | APR   | MAY   |
| Tmax                                                                          | 13 | 0.06      | 0.2   | 0.17  | 0.19       | 0.02  | 0.06  | -0.07  | 0.02  | 0.06  | -0.02 | -0.06 | 0.02  | 0.06         | 0.08  | 0.09  | -0.02            | 0.06  | 0.07  |
|                                                                               | 14 | 0.05      | 0.2   | 0.16  | 0.2        | 0.01  | 0.09  | -0.11  | 0.01  | 0.06  | 0.03  | -0.04 | 0.04  | 0.01         | 0.08  | 0.09  | -0.1             | 0.06  | 0.02  |
|                                                                               | 15 | 0.03      | 0.22  | 0.16  | 0.11       | 0.02  | 0.1   | 0      | 0.1   | 0.09  | 0     | 0     | 0.05  | -0.02        | 0.11  | 0.11  | -0.09            | 0.04  | 0.03  |
|                                                                               | 16 | 0.08      | 0.28  | 0.22  | 0.11       | 0.07  | 0.15  | 0.07   | 0.25  | 0.14  | 0.12  | 0.05  | 0.1   | 0.03         | 0.16  | 0.13  | -0.04            | 0.11  | 0.11  |
|                                                                               | 03 | 0.88      | 1.14  | 0.97  | 0.69       | 0.49  | 0.72  | 1.35   | 1.1   | 0.86  | 0.49  | 0.46  | 0.68  | 0.92         | 0.67  | 0.86  | 1.35             | 1.12  | 1.0   |
|                                                                               | 04 | 0.94      | 1.28  | 0.98  | 0.52       | 0.48  | 0.78  | 1.55   | 1.12  | 0.91  | 1.05  | 0.34  | 0.59  | 0.7          | 0.76  | 0.88  | 1.23             | 1.18  | 1.06  |
| Tmin                                                                          | 05 | 1.23      | 1.35  | 0.95  | 0.93       | 0.87  | 0.9   | 1.63   | 1.26  | 0.92  | 1.04  | 0.56  | 0.82  | 0.95         | 1.03  | 0.94  | 1.29             | 1.25  | 1.09  |
|                                                                               | 06 | 1.27      | 1.26  | 1.03  | 0.47       | 0.53  | 0.84  | 1.34   | 1.32  | 0.94  | 0.87  | 0.76  | 0.95  | 0.99         | 1.09  | 0.94  | 1.26             | 1.29  | 1.12  |
|                                                                               | 13 | -0.85     | -0.6  | 0.05  | -0.6       | -0.55 | -0.49 | -1.07  | -0.84 | -0.77 | -1    | -0.45 | -0.1  | -0.92        | -0.63 | -0.74 | -0.95            | -0.86 | -0.6  |
|                                                                               | 14 | -0.81     | -0.68 | 0.09  | -0.71      | -0.68 | -0.53 | -1.17  | -0.94 | -0.86 | -1.09 | -0.46 | -0.12 | -1.05        | -0.55 | -0.78 | -1.04            | -0.9  | -0.74 |
|                                                                               | 15 | -0.85     | -0.7  | 0.04  | -0.79      | -0.73 | -0.58 | -1.19  | -0.96 | -0.86 | -0.9  | -0.35 | -0.18 | -1.1         | -0.62 | -0.86 | -1.1             | -0.93 | -0.78 |
|                                                                               | 16 | -0.85     | -0.68 | -0.03 | -0.75      | -0.71 | -0.53 | -1.16  | -0.93 | -0.79 | -0.94 | -0.58 | -0.11 | -1.11        | -0.46 | -0.76 | -1.09            | -0.94 | -0.72 |
|                                                                               | 03 | 0.12      | 0.18  | 0.06  | 0.11       | 0.02  | 0.01  | -0.06  | -0.11 | -0.18 | 0.1   | 0.08  | -0.01 | 0.17         | 0.11  | 0.02  | -0.2             | -0.07 | 0.16  |
|                                                                               | 04 | 0.05      | 0.22  | 0.07  | 0.01       | 0.02  | 0.02  | 0.04   | -0.1  | -0.18 | 0.09  | 0.04  | 0.03  | 0.03         | 0.09  | 0.04  | -0.22            | -0.12 | 0.17  |
|                                                                               | 05 | 0.25      | 0.21  | 0.08  | 0.19       | 0.06  | 0.04  | 0.01   | -0.09 | -0.09 | 0.14  | 0.08  | 0.03  | 0.07         | 0.07  | 0.05  | -0.21            | -0.08 | 0.18  |
|                                                                               | 06 | 0.1       | 0.23  | 0.1   | 0.02       | 0.07  | 0.03  | -0.04  | -0.04 | -0.08 | 0.11  | 0.08  | 0.04  | 0.18         | 0.07  | 0.07  | -0.24            | -0.1  | 0.16  |

### 3.4 Heat islands in Pune

After investigating the trend of  $T_{\max}$  and  $T_{\min}$  during the premonsoon season, we explored the phenomenon of heat islands across all observational stations. The manifestation of heat islands serves as a direct indicator of the impacts of urbanization and human activities on the local climate conditions. The climatological trend of UHI was found to be increasing (Figure 4) at Chinchwad, Magarpatta, and Shivajinagar stations from 2012 to 2023. While considering the time variation of temperature, we focused on the premonsoon (MAM) season for our study.

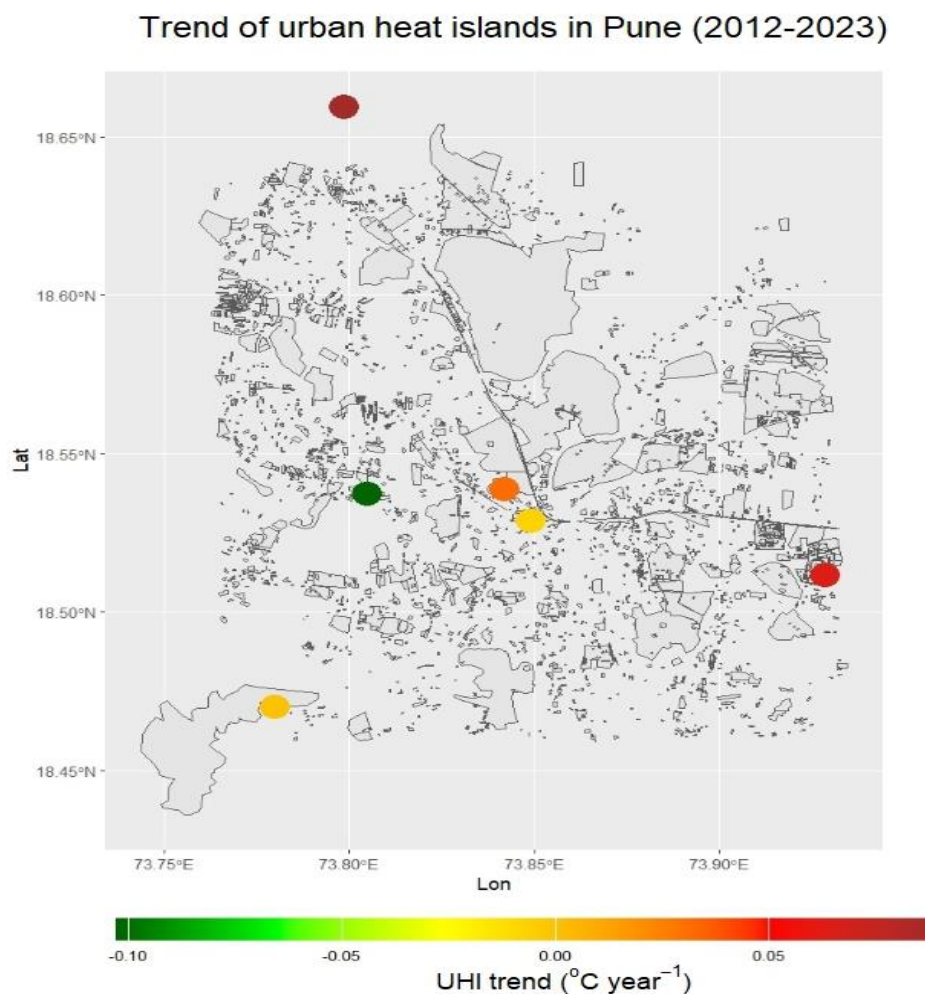


Figure 4. Significant trend of UHI in Pune from 01<sup>st</sup> January 2012 01:00 hour to 31<sup>st</sup> December 2023 23:00 hour.

### 3.4.1 Heat islands in March

Pune has experienced a range of heat islands ( $-1^{\circ}\text{C}$  to  $8^{\circ}\text{C}$ ) in March, from 2012 to 2023, with the Magarpatta station recording the peak UHI temperature during the night (Figure 5). Significant heat islands have also been detected in Chinchwad, Shivajinagar, and Pashan, with UHI temperatures fluctuating between  $2^{\circ}\text{C}$  and  $8^{\circ}\text{C}$  throughout both day and night. In 2018, Pashan recorded the highest UHI temperature during the daytime, particularly at the 13th and 14th hours. The trend of UHI has been noted to decrease (Table 3) in Pashan at night, attributed to the station's closeness to Pashan Lake, which results in a rise in relative humidity and a drop in temperature during nighttime.

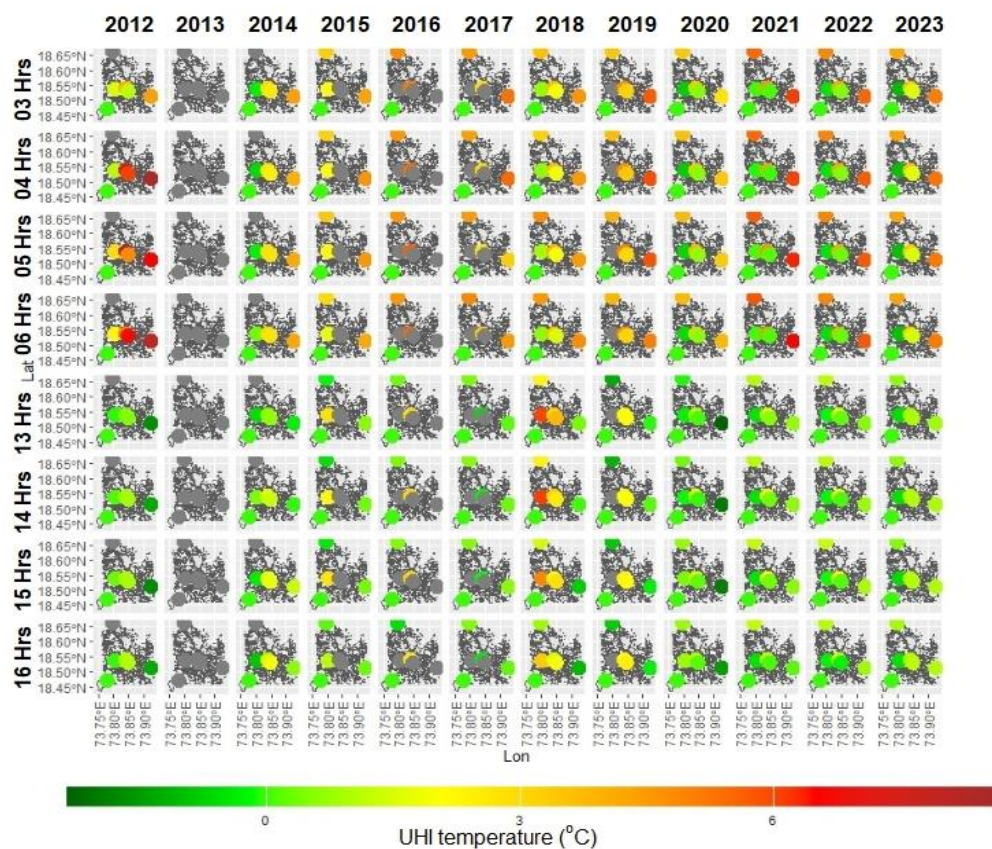


Figure 5. UHI temperature ( $^{\circ}\text{C}$ ) in the study area for the month of March from 2012 to 2023 at daytime and nighttime threshold hours. The color gradient from green (low) to dark red (high) represents the UHI intensity with respect to non-urban (NDA) station.

### 3.4.2 Heat islands in April

Nighttime UHIs in April (Figure 6) are observed at Chinchwad, Shivajinagar, Pashan, and Magarpatta, exhibiting varying intensities ( $2.5^{\circ}\text{C}$  to over  $6^{\circ}\text{C}$ ). Throughout the month of April, Chinchwad and Magarpatta consistently display higher UHI temperatures in night. Conversely, during the daytime, a relatively

minor (below  $1.5^{\circ}\text{C}$ ) variation in UHI temperatures is noted across all locations, with Pashan recording the highest UHI intensity. A notable upward trend in UHI temperature is identified in Chinchwad ( $0.27^{\circ}\text{C year}^{-1}$ ; 14th hour) and Shivajinagar ( $0.13^{\circ}\text{C year}^{-1}$ ; 13th hour) during daytime hours (Table 3).

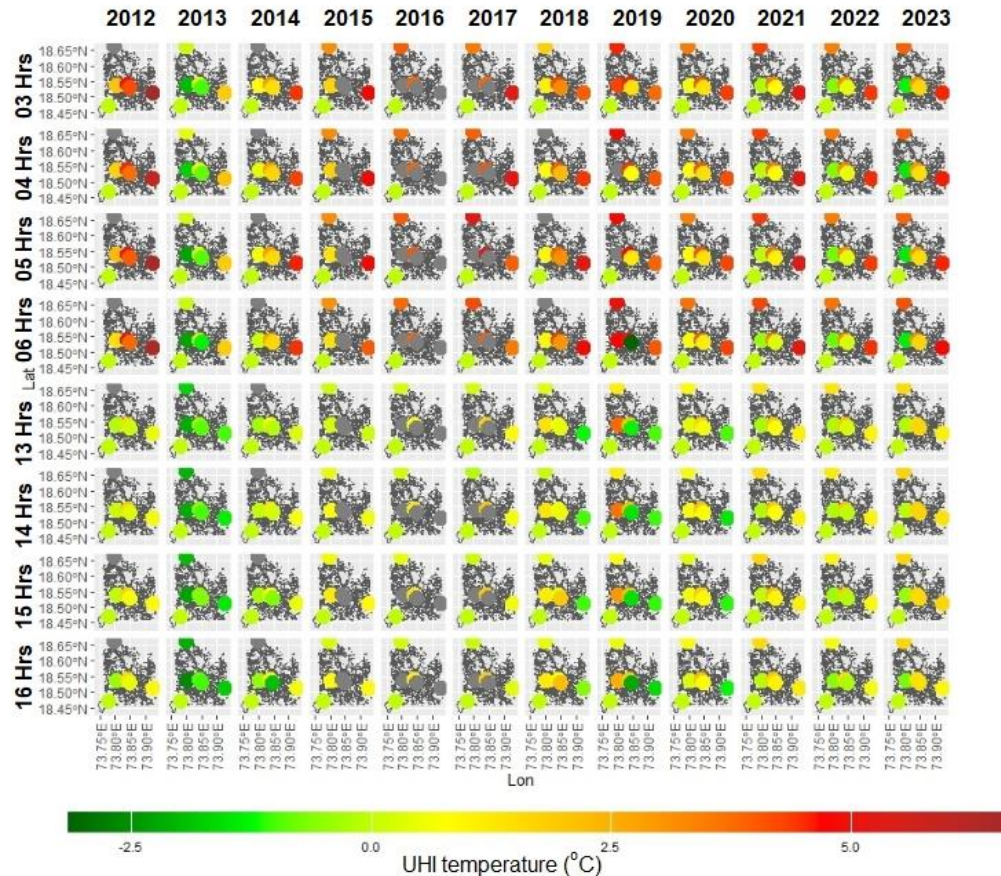


Figure 6. UHI temperature ( $^{\circ}\text{C}$ ) in our study area for the month of April from 2012 to 2023 at daytime and nighttime threshold hours. The color gradient from green (low) to dark red (high) represents the UHI intensity with respect to non-urban (NDA) station.

### 3.4.3 Heat islands in May

The elevated daytime temperatures observed in Pune during May significantly influenced a relatively mild Urban Heat Island (UHI) effect in the area. Since 2013, locations like Magarpatta, Chinchwad, and Shivajinagar have displayed UHIs with varying degrees of intensity, ranging from  $2^{\circ}\text{C}$  to more than  $4^{\circ}\text{C}$  (Figure 7). In 2014, Pashan has shown the highest UHI intensity ( $7.5^{\circ}\text{C}$ ) at 06<sup>th</sup> hour.

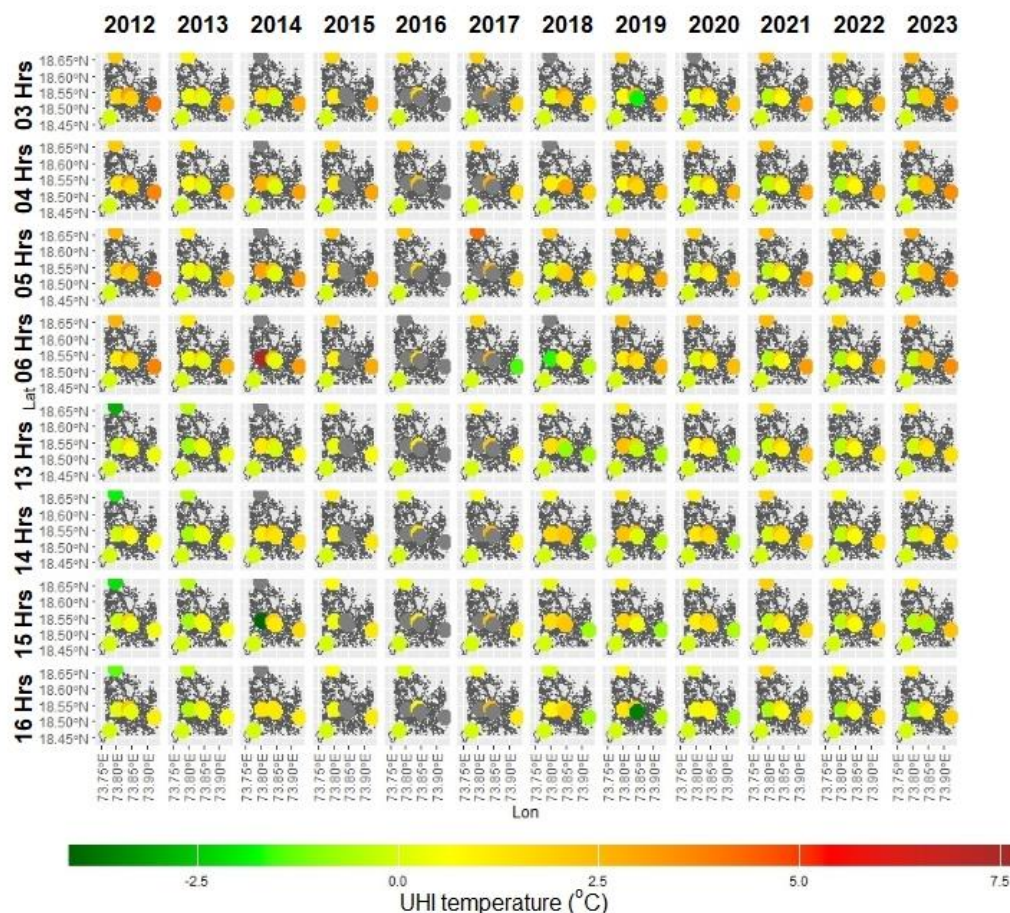


Figure 7. UHI temperature (°C) in the study area for May from 2012 to 2023 at daytime and nighttime threshold hours. The color gradient from green (low) to dark red (high) represents the UHI intensity with respect to non-urban (NDA) station.

Specifically, Chinchwad and Shivajinagar show a clear upward trend in UHI temperatures at different times of day, with Chinchwad peaking at  $0.19^{\circ}\text{C year}^{-1}$  during the 14th hour and Shivajinagar reaching a low of  $0.1^{\circ}\text{C year}^{-1}$  at the 13th hour (Table 3). Nighttime UHIs consistently demonstrate increasing intensities from  $3^{\circ}\text{C}$  to above  $4^{\circ}\text{C}$  in Chinchwad, Magarpatta, and Shivajinagar, although no significant trends have been identified at these sites. Interestingly, Pashan shows a downward trend in UHI temperatures, fluctuating between  $-0.2^{\circ}\text{C year}^{-1}$  and  $-0.13^{\circ}\text{C year}^{-1}$  during the 05th and 06th hours of the day.

Table 3: Trend ( $^{\circ}\text{C year}^{-1}$ ) of UHI temperature at each station from 2012 to 2023 in pre-monsoon season. The highlighted text in red indicates a significant increase/decrease trend.

| Hour      | Chinchwad |             |             | Magarpatta |      |      | Pashan       |       |              | Shivajinagar |             |             | Shivajinagar_ARG |             |            |
|-----------|-----------|-------------|-------------|------------|------|------|--------------|-------|--------------|--------------|-------------|-------------|------------------|-------------|------------|
|           | Mar       | Apr         | May         | Mar        | Apr  | May  | Mar          | Apr   | May          | Mar          | Apr         | May         | Mar              | Apr         | May        |
| <b>13</b> | 0.13      | <b>0.23</b> | <b>0.16</b> | 0.1        | 0.07 | 0.08 | -0.04        | 0.01  | -0.02        | 0.03         | <b>0.08</b> | <b>0.05</b> | -0.02            | <b>0.13</b> | <b>0.1</b> |
| <b>14</b> | 0.12      | <b>0.27</b> | <b>0.19</b> | 0.11       | 0.04 | 0.02 | -0.09        | -0.02 | -0.06        | 0            | 0.05        | 0.05        | -0.1             | 0.09        | 0.02       |
| <b>15</b> | 0.12      | <b>0.23</b> | <b>0.14</b> | 0.08       | 0.05 | 0.11 | -0.08        | -0.01 | -0.02        | 0.01         | 0.08        | 0.05        | -0.09            | 0.07        | -0.01      |
| <b>16</b> | 0.14      | <b>0.22</b> | <b>0.16</b> | 0.09       | 0.03 | 0.06 | -0.05        | -0.01 | -0.11        | 0            | 0.07        | 0.04        | -0.12            | 0.15        | 0.04       |
| <b>03</b> | 0.06      | 0.17        | 0.09        | 0.1        | 0    | 0    | <b>-0.23</b> | -0.25 | -0.16        | -0.01        | -0.08       | -0.04       | -0.09            | -0.14       | 0.04       |
| <b>04</b> | 0.13      | 0.14        | 0.07        | 0.04       | 0.02 | 0    | <b>-0.21</b> | -0.22 | -0.14        | -0.19        | -0.05       | -0.04       | <b>-0.36</b>     | -0.12       | 0.04       |
| <b>05</b> | 0.08      | 0.08        | -0.02       | 0.08       | 0.06 | 0    | <b>-0.33</b> | -0.24 | <b>-0.2</b>  | -0.18        | -0.12       | 0           | <b>-0.35</b>     | -0.12       | 0.04       |
| <b>06</b> | 0.11      | 0.14        | 0.07        | 0.11       | 0.09 | 0    | <b>-0.28</b> | -0.16 | <b>-0.13</b> | -0.04        | -0.08       | -0.03       | <b>0.39</b>      | -0.15       | 0.07       |

The identification of the predominant land cover class at each specific station was carried out through the application of a supervised machine learning method. This involved the development of a multivariate linear regression model for UHI temperature, incorporating land cover classes as independent

variables. The data was organized on the basis of categorical proportions of land cover classes in the surrounding 1 km<sup>2</sup> area at each station. The statistical evaluation of the model indicated the significance of the relationships between land cover classes, with  $r^2$  values ranging from 0.001 to 0.013 and  $P$ -values ranging from  $7.91 \times 10^{-13}$  to  $2.33 \times 10^{-5}$  across the study locations. The strong individual relationship with UHI was underscored by the model's significance and the lower  $P$ -values. An analysis of variance (ANOVA) test was performed to validate the model's statistics. ANOVA divides the total variability observed in a dataset into systematic and random factors. Unlike random factors, systematic factors have a statistically significant impact on the dataset. ANOVA, also referred to as Fisher's analysis of variance, extends beyond  $t$ - and  $Z$ -tests. The test outcomes confirmed the significance of the model through the  $F$ -value for each estimator. Detailed model and ANOVA statistics are outlined in Table 4.

Table 4: Statistical analysis results of a multivariate linear regression model of UHI with the variables cropland, herbaceous cropland, irrigated cropland, mosaic cropland, natural Vegetation, Vegetation, grassland and urban, and their coefficients of regression ( $\beta$ ), along with an analysis of variance. Results include standard error (SE),  $t$ -value ( $t$ ), sum of squares (SS), mean of squares (MS), and  $F$ -value ( $F$ ).

| Modeled statistics for land cover classes |                       |                                 |       |       |                      |       |       |
|-------------------------------------------|-----------------------|---------------------------------|-------|-------|----------------------|-------|-------|
|                                           |                       | Individual linear relationships |       |       | Analysis of variance |       |       |
| Station                                   | Dominating land cover | $\beta$                         | SE    | $t$   | SS                   | MS    | $F$   |
| Chinchwad                                 | Urban                 | 0.021                           | 0.01  | 1.116 | 4.05                 | 4.054 | 1.245 |
| Magarpatta                                | Urban                 | 0.011                           | 0.023 | 0.5   | 1.38                 | 1.380 | 0.25  |
| Pashan                                    | Mixed                 | 0.26                            | 0.107 | 2.5   | 447.4                | 2.27  | NA    |
| Shivajinagar                              | Natural Vegetation    | 0.568                           | 0.124 | 4.54  | 36.07                | 36.06 | 18.6  |
|                                           | Urban                 | 0.025                           | 0.017 | 1.42  | 3.91                 | 3.91  | 2.02  |
| Shivajinagar_ARG                          | Cropland              | 0.027                           | 0.04  | 0.61  | 0.74                 | 0.74  | 0.38  |

Chinchwad and Magarpatta have demonstrated the influence of the urban environment on UHI intensities throughout the study period. On the other hand, Shivajinagar has reported the dominance of cropland, natural vegetation, and urban environment, while Pashan has shown the influence of mixed land cover on UHI. The combined effect of temperature and RH on UHI in Pune has revealed that at the 04<sup>th</sup>, 05<sup>th</sup>, and 06<sup>th</sup> hour, UHI intensity is highest when near-surface temperature ranges from 20°C to 22°C and RH is from 40% to 60%. At the 03<sup>rd</sup> hour, elevated UHI temperature is observed at higher near-surface temperature (22°C to 26°C) and RH values between 50% to 70% (Fig. 8). Conversely, in daytime, UHIs are mostly formed when near-surface temperature increases beyond 35°C in the presence of low humid conditions (below 35%).

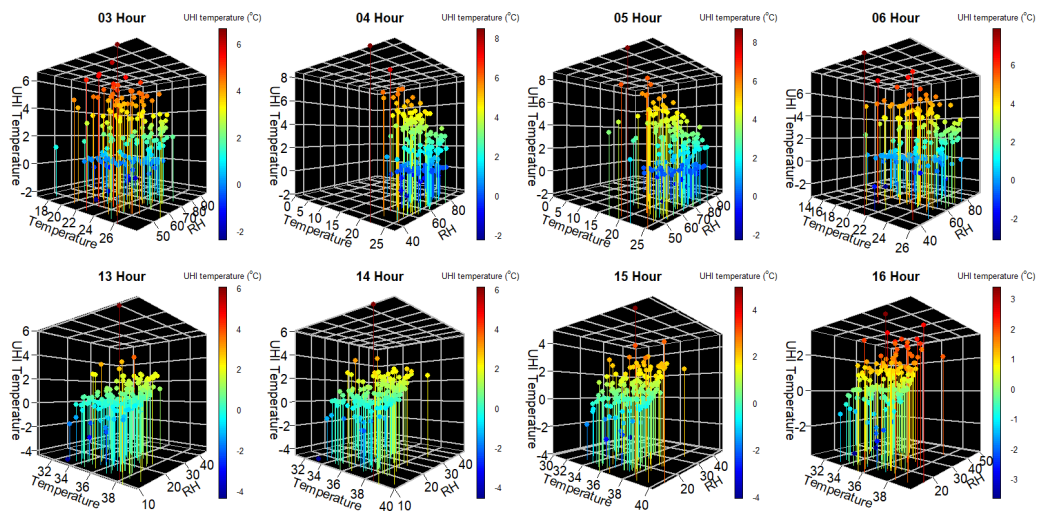


Figure 8. 3D scatter plot of UHI (Z-axis) with temperature (X-axis) and RH (Y-axis) in Pune at night time (upper panels) and day time (lower panels) threshold hours from 2012 to 2023. The color gradient from blue to red indicates the UHI intensity ( $^{\circ}\text{C}$ ) from low (blue) to high (red).

#### 4. Conclusion

In the current work, we investigated the impact of microclimatic conditions and urbanization on heat islands in India's rapidly growing Pune city. As far as our knowledge goes, very few studies have reported the UHI effect within Pune city. In the present work, we evaluated the impact of land cover and atmospheric conditions on the sub-urban of Pune city and ultimately determined that land cover-induced atmospheric instability evolved the day and night heat islands within the city with magnitudes up to  $8^{\circ}\text{C}$ . Our results demonstrated that increasing maximum and minimum temperature trends significantly subsidized the atmospheric imbalances. Utilizing a supervised machine learning algorithm, we quantified the dominating land cover type in different locations of Pune and have reported that urban infrastructure contributed to UHIs in Chinchwad and Magarpatta regions while mixed land cover types under the influence of ideal meteorological conditions contributed to UHIs in Pashan and Shivajinagar areas. We also investigated the effect of local meteorological parameters on UHIs to highlight the specific conditions during the day and night that intensified UHI effects. Our findings reveal that high daytime temperature (above  $35^{\circ}\text{C}$ ) and low relative humidity (below 35%) favored daytime UHIs, whereas nighttime UHIs were influenced by low temperatures ( $20^{\circ}\text{C}$  to  $26^{\circ}\text{C}$ ) and moderate humidity levels (40%-70%). The study has shown that the evolution of heat islands resulting from changes in land cover can be efficiently achieved by applying a machine learning framework to the most current of land use datasets. However, more accurate fine-scale analyses require improved data on more meteorological parameters and high-resolution LULC data with an expanded range of sub-categories. Numerous research endeavors are underway to detect hot pockets early, and machine learning algorithms have proven to be

particularly effective in monitoring UHIs. The methodology presented in our paper will contribute to ongoing research efforts related to UHI monitoring and impact assessment, both in India and globally.

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